

Media_Streaming_Service Database Relational Schema

User = (AccountID, Name, Email, Bdate)

Content = (ContentID, Title, Release_date, Maturity, Length, Language)

- Maturity $\in$ {'Explicit', 'Not Explicit'}

Subscription = (SubscriptionID, Start_Date, End_Date, Cost, Subscription_type, Tier, Max_Family_size)

- Tier $\in$ {'Premium', 'Deluxe'}
- Subscription_Type $\in$ {'Individual', 'Family'}

Listener = (ListenerID[FK1], Username, Timestamp, Content[FK7], Enroll[FK8])

- FK1: ListenerID $\rightarrow$ User(AccountID)
 - **On update cascade:** If AccountID is updated then this must be updated in the Listener table too
 - **On delete cascade:** If the User's account is deleted then the User's listener profile must be deleted
- FK7: Content $\rightarrow$ Content(ContentID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete set null:** If content is deleted then the listener stops streaming that content, but the listener's account remains active
- FK8: Enroll $\rightarrow$ Subscription(SubscriptionID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete set null:** A listener does not have to enroll in a subscription

Creator = (CreatorID[FK2], Stage_name, Biography, Pin[FK9])

- FK2: CreatorID $\rightarrow$ User(AccountID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** A creator cannot exist without a user
- FK9: Pin $\rightarrow$ Content(ContentID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete set null:** It is optional to pin content

Socials = (CreatorID[FK3], handle, platform)

- FK3: CreatorID $\rightarrow$ Creator(CreatorID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** Socials cannot exist without a creator

Album = (Name, CreatorID[FK13], AlbumID[FK11])

- FK13: CreatorID $\rightarrow$ Creator(CreatorID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** An album can not exist without a creator
- FK11: AlbumID $\rightarrow$ Content(ContentID)

- **On update cascade:** All IDs must be changed if referenced ID is changed
- **On delete set null:** Creator doesn't have to own an album

Song = (SongID[FK4], AlbumID[FK12])

- FK4: SongID → Content(ContentID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If content is deleted then song must be deleted
- FK12: AlbumID → Album(AlbumID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete set null:** Songs don't have to belong to an album

Genres = (SongID[FK5], genre_name)

- FK5: SongID → Song(SongID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** Genres cannot exist without a song

Podcast_Series = (PodcastID, Description, Title)

Podcast_Episode = (EpID[FK6], Topic, Episode_number, PodcastID[FK10])

- FK6: EpID → Content(ContentID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If the content is deleted then the podcast episode (type of content) would also be deleted
- FK10: PodcastID → Podcast_series(PodcastID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If the podcast ID is deleted then the podcast episode, since it makes up the podcast, would also be deleted

Playlist = (PlaylistID, Name, ListenerID[FK18])

- FK18: ListenerID → Listener(ListenerID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete set null:** if the listener is deleted then the playlist can continue to exist without a listenerID being connected

Creates = (CreatorID[FK14], ContentID[FK15])

- FK14: CreatorID → Creator(CreatorID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If creator is deleted then the creates table input that connects the creator with the content they create will also be deleted
- FK15: ContentID → Content(ContentID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If the content is deleted then the connection between the content and the creator who created it would also be deleted

Makes_up = (PlaylistID[FK16], SongID[FK17], Track_Order)

- FK16: PlaylistID → Playlist(PlaylistID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If the playlist is deleted then no songs will make up the playlist so the connection will be deleted
- FK17: SongID → Song(SongID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If the songID for the song in the playlist is deleted then that songs connection to the playlist will also be deleted

Friends = (Send[FK19], Receive[FK20])

- FK19: Send → Listener(LisenterID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If a listener is deleted then connection to their friends will also be deleted
- FK20: Receive → Listener(ListenerID)
 - **On update cascade:** All IDs must be changed if referenced ID is changed
 - **On delete cascade:** If a listener is deleted then connection to their friends will also be deleted